Techno-Economic Assessment of **Steel Slag** Valorization

H_2 Production, CO_2 Sequestration, and Iron Recovery

Step Function LLC

Separation-Agents Framework

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Abstract

This study presents a techno-economic assessment of valorizing basic oxygen furnace (BOF) steel slag through three integrated pathways: hydrogen production via serpentinization, CO_2 sequestration via mineral carbonation, and iron concentrate recovery via magnetic separation. A Generalized Disjunctive Programming (GDP) superstructure with 4 feasible topologies was evaluated at 500 t/d throughput using the Separation-Agents framework. Reaktoro thermodynamic speciation informed the equilibrium constraints, and JAX-based techno-economic analysis (TEA) ranked all configurations. The optimal topology (heat recovery only, no inter-reactor scrubber) yields a net value of **\$37.87/t** feedstock (\$6.25M/yr), driven by iron concentrate revenue (\$5.05M/yr), CO_2 credits (\$3.51M/yr), and green hydrogen (\$0.96M/yr).

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1 Introduction and Opportunity

Basic oxygen furnace (BOF) steel slag is a high-volume industrial byproduct generated at rates of 100–200 kg per tonne of crude steel. Global steel production exceeds 1.8×10^9 t/yr, yielding over 250×10^6 t/yr of BOF slag [1]. While some slag is recycled as construction aggregate, the majority is landfilled, representing both an environmental liability and a missed resource opportunity.

BOF slag's mineral composition—rich in CaO, FeO/Fe_2O_3, MgO, and SiO_2—makes it an attractive feedstock for:

1. **Green hydrogen** via serpentinization of fayalite (Fe_2SiO_4),
2. **CO_2 sequestration** via mineral carbonation of CaO and MgO, and
3. **Iron concentrate recovery** via low-intensity magnetic separation (LIMS).

This study applies the Separation-Agents GDP optimization framework to identify the process topology that maximizes net economic value per tonne of slag.

2 Raw Material Composition

The BOF steel slag composition assumed in this study is representative of typical converter slag from integrated steelworks.

Table 1: BOF steel slag composition (wt%).

Component	**wt%**	**Role in Valorization**
CaO	40	CO_2 mineralization $\rightarrow$ $CaCO_3$
Fe_2O_3	15	Iron concentrate; oxide feed
FeO	15	Fayalite source for H_2; iron recovery
SiO_2	20	Gangue (reports to quartz)
MgO	10	CO_2 mineralization $\rightarrow$ $MgCO_3$

3 Process Description

The process comprises three sequential stages, informed by Reaktoro equilibrium speciation results showing that iron removal is required before carbonation to maintain a pH favorable for carbonate precipitation.

3.1 Process Flow Diagram

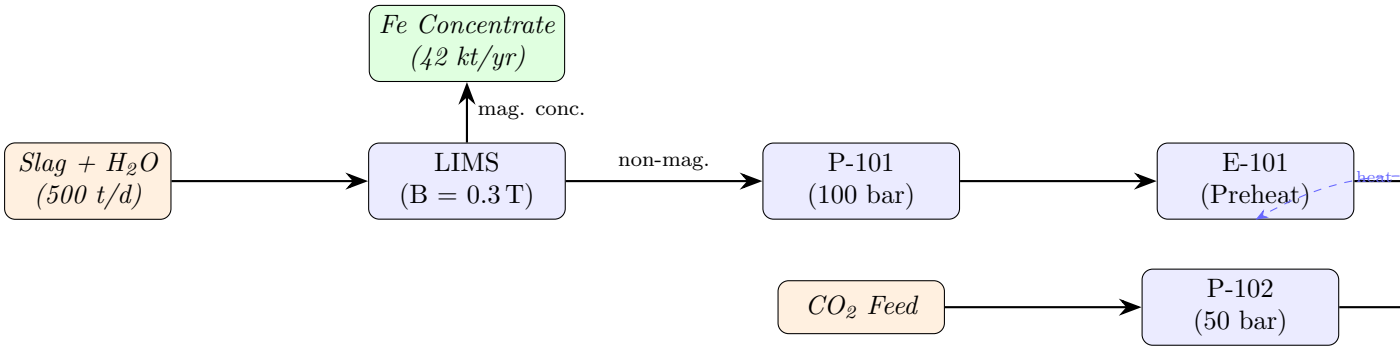

Figure 1: Process flow diagram for steel slag valorization (optimal topology: heat recovery only).

3.2 Stage 1: Magnetic Separation

Low-intensity magnetic separation (LIMS) recovers magnetite (Fe_3O_4) and hematite from the slag at 85% recovery. The iron concentrate (approx. 42,000 t/yr at 500 t/d) is sold directly.

3.3 Stage 2: Serpentinization — H_2 Production

The non-magnetic fraction, rich in fayalite (Fe_2SiO_4), undergoes serpentinization:

$$Fe_2SiO_4 + 3\,H_2O \longrightarrow 2\,Fe(OH)_2 + SiO_2 + H_2 \tag{1}$$

Conditions: 250 °C, 100 bar, 4 h residence time, 70% conversion. Reaktoro speciation at these conditions predicts pH 5.6 and formation of magnetite as a secondary product.

3.4 Stage 3: Mineral Carbonation — CO_2 Sequestration

The liquid effluent from serpentinization (now depleted in iron, raising the pH) contacts CO_2 at 185 °C and 50 bar:

$$CaO + CO_2 \longrightarrow CaCO_3 \tag{2}$$

$$MgO + CO_2 \longrightarrow MgCO_3 \tag{3}$$

At 80–85% conversion, approximately 58,575 t CO_2 per year is permanently sequestered as stable carbonate minerals.

4 Modeling Approach

Table 2: Three-tier modeling hierarchy.

Tier	Tool	Role
Thermodynamic	Reaktoro (SUPCRTBL)	Gibbs energy minimization for aqueous speciation, mineral saturation indices, pH and Eh prediction
Process	IDAES-PSE + Pyomo.GDP	Sequential-modular/equation-oriented flowsheet simulation; GDP topology optimization
Economic	JAX TEA	Differentiable cost modeling (CAPEX/OPEX); autodiff-based sensitivity analysis

4.1 Thermodynamic Speciation

Three Reaktoro equilibrium scenarios were evaluated:

Table 3: Reaktoro speciation results for steel slag.

Scenario	pH	E_h (V)	Key Finding
Ambient (25°C, 1 atm)	2.43	0.656	Fe^{3+} hydrolysis dominates; highly acidic
Serpentinization (250°C, 100 bar)	5.60	0.661	0.7% fayalite equilibrium conversion; H_2(aq) produced
Carbonation (150°C, 50 bar, +CO_2)	3.20	0.106	<0.1% CO_2 conversion at low pH

Critical insight: Iron removal must precede carbonation, as dissolved Fe^{3+} suppresses pH to ∼2–3, thermodynamically preventing carbonate precipitation.

4.2 GDP Superstructure

The `steel_slag_h2_co2` superstructure contains 15 unit operations (12 fixed, 3 optional) with:

- 1 disjunction: auxiliary heater `XOR` heat-recovery-only
- 1 optional unit: inter-reactor scrubber
- 4 feasible topologies
- 12 continuous design variables (reactor T, P, τ; HX areas; pump heads)

5 Cost Model Assumptions

Table 4: Plant-level parameters.

Parameter	**Value**
Slag throughput	500 t/d
Operating days	330 d/yr
Annual throughput	165,000 t/yr
Discount rate	8%
Project lifetime	20 yr

Table 5: Commodity prices.

Product	**Price**	**Source**
H_2 (green)	\$4/kg	DOE target
CO_2 credit	\$60/t	Voluntary carbon market
Fe concentrate	\$120/t	FOB spot

6 Results

6.1 Topology Ranking

Table 6: Trade-study results ranked by net value.

Rank	**Topology**	**CAPEX** (\$M)	**OPEX** (\$M/yr)	**EAC** (\$M/yr)	**Revenue** (\$M/yr)	**Net Value** (\$M/yr)
1	Heat recovery only	7.76	2.49	3.28	9.53	**6.25**
2	HR + scrubber	7.96	2.49	3.30	9.53	6.23
3	Aux heater	8.06	3.07	3.89	9.53	5.64
4	Aux heater + scrubber	8.26	3.07	3.91	9.53	5.62

6.2 Revenue Breakdown

Table 7: Annual revenue by product stream (optimal topology).

Product	**Production**	**Price**	**Revenue (\$M/yr)**
Fe concentrate	42,075 t/yr	\$120/t	5.05
CO_2 credits	58,575 t/yr	\$60/t CO_2	3.51
H_2 gas	241 t/yr	\$4/kg	0.96
Total			**9.53**

6.3 Levelized Economics

Table 8: Levelized economics for the optimal topology (Config 3).

Metric	**Value**
Total CAPEX	\$7.76M
Annual OPEX	\$2.49M/yr
EAC (8%, 20 yr)	\$3.28M/yr
Total Revenue	\$9.53M/yr
Net Value	**\$6.25M/yr**
Net Value/t	**\$37.87/t**

6.4 Cost Sensitivity

JAX autodifferentiation identified the top cost drivers (∂EAC/∂parameter):

Table 9: Top-5 cost sensitivity parameters (JAX autodiff).

Parameter	$\lvert\partial\mathbf{EAC}/\partial x\rvert$	**Interpretation**
Residence time (h)	\$10,185/h	Reactor sizing driver
Operating temp (°C)	\$8,250/°C	Energy cost driver
Throughput (t/d)	\$5,214/t/d	Economies of scale

7 Discussion

7.1 Economic Drivers

Iron concentrate dominates revenue at 53% of total, followed by CO_2 credits (37%) and H_2 (10%). The marginal value of the auxiliary heater (−\$0.61M/yr net) does not justify the additional \$0.30M CAPEX and \$0.58M/yr heating OPEX. Similarly, the inter-reactor scrubber provides minimal benefit (−\$0.02M/yr).

7.2 Key Risks

- **CO_2 credit volatility**: At \$30/t (instead of \$60/t), CO_2 revenue halves to \$1.76M/yr, reducing net value to \$4.49M/yr (\$27.22/t).
- **Serpentinization kinetics**: Reaktoro predicts only 0.7% equilibrium conversion in batch; achieving 70% requires extended residence time and catalytic enhancement.
- **Iron price sensitivity**: Fe concentrate at \$80/t (vs. \$120/t) would reduce revenue by \$1.68M/yr.

7.3 Modeling Limitations

1. Reaktoro speciation uses the SUPCRTBL database, which lacks Cr, Mn, and V species commonly present in BOF slag.
2. Serpentinization kinetics are not modeled; the 70% conversion is assumed based on literature values for autoclave reactors.
3. The JAX TEA model uses simplified cost correlations; detailed vendor quotes would be needed for a bankable feasibility study.

8 Conclusions

1. Steel slag valorization at 500 t/d is economically attractive, yielding a net value of **\$37.87/t** feedstock (\$6.25M/yr).
2. The optimal topology (heat recovery only, no scrubber) minimizes cost while maintaining all three revenue streams.
3. Iron concentrate is the dominant value driver (53% of revenue), followed by CO_2 credits (37%) and H_2 (10%).
4. Iron removal before carbonation is thermodynamically essential (Reaktoro-confirmed).
5. The auxiliary heater and scrubber do not justify their added cost; heat recovery from product streams is sufficient.

References

[1] World Steel Association, *Steel Statistical Yearbook 2023*. Brussels, 2023.

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